

# TSUNAMI AND COASTAL HAZARD MANAGEMENT

TSUNAMI GENERATION CHAIN → COASTAL-HAZARD FIREWALL → SOURCE-TO-IMPACT MATRIX → DETECTION-CONFIRMATION RAIL → ITEWC AUTHORITY MAP → WARNING-UNCERTAINTY LADDER

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • source generation g3 and all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

00

STAGE 00

## TSUNAMI GENERATION CHAIN

TSUNAMI GENERATION CHAIN

SUDDEN WATER DISPLACEMENT

VOLCANIC ACTIVITY

LONG WAVES

COASTAL TRANSFORMATION

RUN-UP + INUNDATION + RETURN FLOW

### RISK CONDITION / INPUT

- SUDDEN WATER DISPLACEMENT
- VOLCANIC ACTIVITY

### PREVENTION / PREPAREDNESS / RESPONSE MECHANISM

- LONG WAVES → COASTAL TRANSFORMATION
- RUN-UP + INUNDATION + RETURN FLOW

### LAST-MILE / INCLUSION / OUTCOME LIMIT

- SUDDEN WATER DISPLACEMENT
- VOLCANIC ACTIVITY

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: LONG WAVES → COASTAL TRANSFORMATION; RUN-UP + INUNDATION + RETURN FLOW.

01

STAGE 01

## COASTAL-HAZARD FIREWALL

COASTAL-HAZARD FIREWALL

SUDDEN DISPLACEMENT

GRAVITATIONAL CYCLE

STORM SURGE

CYCLONE WIND + PRESSURE

DIFFERENT MONITORING AND RESPONSE LOGIC

### HAZARD / ACTOR / INSTRUMENT

TSUNAMI → sudden displacement  
TIDE → gravitational cycle

### DECISIVE RISK OR MANDATE DISTINCTION

STORM SURGE → cyclone wind + pressure  
RULE → different monitoring and response logic

### WARNING / IMPLEMENTATION / OUTCOME LIMIT

TSUNAMI → sudden displacement  
TIDE → gravitational cycle

### HAZARD / ACTOR / INSTRUMENT

- TSUNAMI → sudden displacement
- TIDE → gravitational cycle

### DECISIVE RISK OR MANDATE DISTINCTION

- STORM SURGE → cyclone wind + pressure
- RULE → different monitoring and response logic

### WARNING / IMPLEMENTATION / OUTCOME LIMIT

- TSUNAMI → sudden displacement
- TIDE → gravitational cycle

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: STORM SURGE → cyclone wind + pressure; RULE → different monitoring and response logic.

02

STAGE 02

## SOURCE-TO-IMPACT MATRIX

SOURCE-TO-IMPACT MATRIX

MECHANISM DEPTH DISPLACEMENT

DISTANCE AND OCEAN CONFIGURATION

BATHYMETRY TOPOGRAPHY LAND USE

EXPOSURE VULNERABILITY AND ACTION

### HAZARD / ACTOR / INSTRUMENT

SOURCE → mechanism depth displacement  
PATH → distance and ocean configuration

### DECISIVE RISK OR MANDATE DISTINCTION

COAST → bathymetry topography land use  
OUTCOME → exposure vulnerability and action

### WARNING / IMPLEMENTATION / OUTCOME LIMIT

SOURCE → mechanism depth displacement  
PATH → distance and ocean configuration

### HAZARD / ACTOR / INSTRUMENT

- SOURCE → mechanism depth displacement
- PATH → distance and ocean configuration

### DECISIVE RISK OR MANDATE DISTINCTION

- COAST → bathymetry topography land use
- OUTCOME → exposure vulnerability and action

### WARNING / IMPLEMENTATION / OUTCOME LIMIT

- SOURCE → mechanism depth displacement
- PATH → distance and ocean configuration

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: COAST → bathymetry topography land use; OUTCOME → exposure vulnerability and action.

03

STAGE 03

## DETECTION-CONFIRMATION RAIL

DETECTION-CONFIRMATION RAIL

1 SEISMIC EVENT ANALYSIS

2 PRE-RUN MODEL SCENARIO

3 SEA-LEVEL MONITORING

4 UPDATED BULLETIN

THREAT STATUS

### HAZARD / RISK / RESILIENCE CONCEPT

- 1 SEISMIC EVENT ANALYSIS
- 2 PRE-RUN MODEL SCENARIO

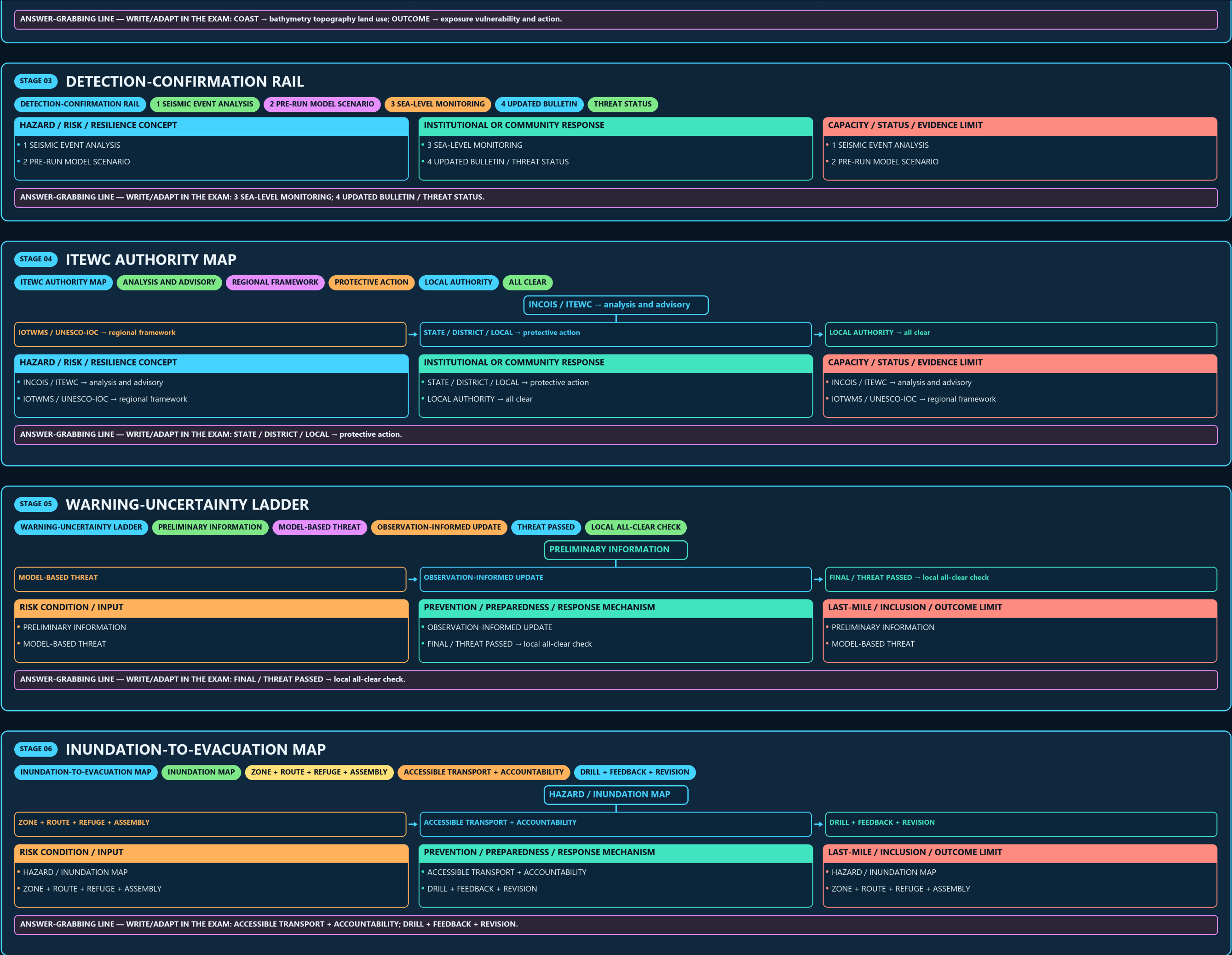
### INSTITUTIONAL OR COMMUNITY RESPONSE

- 3 SEA-LEVEL MONITORING
- 4 UPDATED BULLETIN / THREAT STATUS

### CAPACITY / STATUS / EVIDENCE LIMIT

- 1 SEISMIC EVENT ANALYSIS
- 2 PRE-RUN MODEL SCENARIO

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: 3 SEA-LEVEL MONITORING; 4 UPDATED BULLETIN / THREAT STATUS.



06

STAGE 06 INUNDATION-TO-EVACUATION MAP

INUNDATION-TO-EVACUATION MAP

INUNDATION MAP

ZONE + ROUTE + REFUGE + ASSEMBLY

ACCESSIBLE TRANSPORT + ACCOUNTABILITY

DRILL + FEEDBACK + REVISION

HAZARD / INUNDATION MAP

ZONE + ROUTE + REFUGE + ASSEMBLY

ACCESSIBLE TRANSPORT + ACCOUNTABILITY

DRILL + FEEDBACK + REVISION

RISK CONDITION / INPUT

- HAZARD / INUNDATION MAP
- ZONE + ROUTE + REFUGE + ASSEMBLY

PREVENTION / PREPAREDNESS / RESPONSE MECHANISM

- ACCESSIBLE TRANSPORT + ACCOUNTABILITY
- DRILL + FEEDBACK + REVISION

LAST-MILE / INCLUSION / OUTCOME LIMIT

- HAZARD / INUNDATION MAP
- ZONE + ROUTE + REFUGE + ASSEMBLY

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: ACCESSIBLE TRANSPORT + ACCOUNTABILITY; DRILL + FEEDBACK + REVISION.

07

STAGE 07 COASTAL PROTECTION PORTFOLIO

COASTAL PROTECTION PORTFOLIO

SHELTERS WALLS BREAKWATERS WHERE JUSTIFIED

DUNES MANGROVES FORESTS REEFS

CRZ + RISK-SENSITIVE SITING

LAYERED AND LOCATION-SPECIFIC

HAZARD / ACTOR / INSTRUMENT

ENGINEERED → shelters walls breakwaters where justified  
NATURAL → dunes mangroves forests reefs

DECISIVE RISK OR MANDATE DISTINCTION

REGULATORY → CRZ + risk-sensitive siting  
RULE → layered and location-specific

WARNING / IMPLEMENTATION / OUTCOME LIMIT

ENGINEERED → shelters walls breakwaters where justified  
NATURAL → dunes mangroves forests reefs

HAZARD / ACTOR / INSTRUMENT

- ENGINEERED → shelters walls breakwaters where justified
- NATURAL → dunes mangroves forests reefs

DECISIVE RISK OR MANDATE DISTINCTION

- REGULATORY → CRZ + risk-sensitive siting
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WARNING / IMPLEMENTATION / OUTCOME LIMIT

- ENGINEERED → shelters walls breakwaters where justified
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ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: REGULATORY → CRZ + risk-sensitive siting; RULE → layered and location-specific.

08

STAGE 08 COASTAL ECONOMY CONTINUITY

COASTAL ECONOMY CONTINUITY

SHUTDOWN ACCESS REOPENING

BOATS LANDING CENTRES MARKETS

VISITOR WARNING SIGNAGE ROUTE

CONTINUITY AND SAFE RECOVERY

PORT / HARBOUR → shutdown access reopening

FISHERIES → boats landing centres markets

TOURISM → visitor warning signage route

UTILITIES / HOSPITAL → continuity and safe recovery

HAZARD / RISK / RESILIENCE CONCEPT

- PORT / HARBOUR → shutdown access reopening
- FISHERIES → boats landing centres markets

INSTITUTIONAL OR COMMUNITY RESPONSE

- TOURISM → visitor warning signage route
- UTILITIES / HOSPITAL → continuity and safe recovery

CAPACITY / STATUS / EVIDENCE LIMIT

- PORT / HARBOUR → shutdown access reopening
- FISHERIES → boats landing centres markets

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: UTILITIES / HOSPITAL → continuity and safe recovery.

09

STAGE 09 TSUNAMI READY CYCLE

TSUNAMI READY CYCLE

MAP RISK

EVACUATION MAP + SIGNAGE

EVALUATE INDICATORS

RENEW READINESS

COMMUNITY RECOGNITION

NOT NATIONAL CERTIFICATION

MUST REMEMBER

MAP RISK → EDUCATE → WARN

EVACUATION MAP + SIGNAGE → DRILL

EVALUATE INDICATORS → RENEW READINESS

COMMUNITY RECOGNITION → NOT NATIONAL CERTIFICATION

MUST REMEMBER: Tsunami risk management links offshore detection, seismic and sea-level

RISK CONDITION / INPUT

- MAP RISK → EDUCATE → WARN
- EVACUATION MAP + SIGNAGE → DRILL

PREVENTION / PREPAREDNESS / RESPONSE MECHANISM

- EVALUATE INDICATORS → RENEW READINESS
- COMMUNITY RECOGNITION → NOT NATIONAL CERTIFICATION

LAST-MILE / INCLUSION / OUTCOME LIMIT

- MUST REMEMBER: Tsunami risk management links offshore detection, seismic and sea-level...
- MAP RISK → EDUCATE → WARN

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MUST REMEMBER: Tsunami risk management links offshore detection, seismic and sea-level.

